

Mathematical Theory Compendium

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1 Overview

This note is an attempt to write broad areas of mathematics through compact and reusable equations. The emphasis is not on encyclopedic completeness. The goal is to show a style of mathematical organization: define the objects cleanly, state the structural laws precisely, and then express higher theories as controlled extensions of the same notation.

The three main threads here are:

- algebra as the study of operations and the laws they satisfy,
- analysis as the study of functions, variation, and differential equations,
- geometry as the study of smooth spaces, tensors, and curvature.

2 Algebra

2.1 Terms and Operations

Let

$$x_i \in X$$

be symbols in a set X . Let

$$f_i : X^{n_i} \rightarrow X$$

be an operation of arity n_i . In prefix notation,

$$f_i x_1 \cdots x_{n_i} = f_i(x_1, \dots, x_{n_i}).$$

This viewpoint is useful because it makes the algebraic laws look like equational templates. A structure is then determined by the operations it admits together with the identities they satisfy.

2.2 Fundamental Algebraic Laws

Closure.

$$f(x, y) \in X.$$

Commutativity.

$$f(x, y) = f(y, x).$$

Associativity.

$$f(f(x, y), z) = f(x, f(y, z)).$$

Identity. There exists $e \in X$ such that

$$f(e, x) = x, \quad f(x, e) = x.$$

Inverse. For every $x \in X$, there exists x^{-1} such that

$$f(x, x^{-1}) = e.$$

Idempotence.

$$f(x, x) = x.$$

Absorption. For two operations f and g ,

$$f(x, g(x, y)) = x.$$

Distributivity. For two operations f and g ,

$$f(x, g(y, z)) = g(f(x, y), f(x, z)).$$

2.3 Basic Algebraic Structures

Magma. A magma is a set X equipped with one binary operation

$$f : X \times X \rightarrow X.$$

Semigroup. A semigroup is a magma satisfying associativity.

Monoid. A monoid is a semigroup with an identity element.

Group. A group is a monoid in which every element has an inverse.

Abelian group. An Abelian group is a group that is also commutative.

These structures form a natural ladder: each new notion is obtained by adding one more law to the previous one.

2.4 Rings and Fields

A ring R has two operations $+$ and $\cdot$ such that

$$(R, +)$$

is an Abelian group,

$$(R, \cdot)$$

is a monoid, and multiplication distributes over addition:

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc.$$

A field $\mathbb{F}$ is a commutative ring for which

$$(\mathbb{F} \setminus \{0\}, \cdot)$$

is an Abelian group. In practice, the field condition is what allows division by nonzero scalars and therefore makes linear theory behave cleanly.

2.5 Modules and Vector Spaces

Let R be a ring and M an Abelian group. A module structure is a scalar multiplication

$$R \times M \rightarrow M, \quad (r, v) \mapsto rv,$$

satisfying

$$\begin{aligned} r(u + v) &= ru + rv, \\ (r + s)v &= rv + sv, \\ (rs)v &= r(sv). \end{aligned}$$

When the base ring is a field $\mathbb{F}$, the module is a vector space. This is the point where algebra becomes linear algebra: vector addition and scalar multiplication are governed by the same distributive pattern, but now over a field rather than an arbitrary ring.

2.6 Associative and Lie Algebras

An associative algebra over a field $\mathbb{F}$ is a vector space A equipped with a bilinear multiplication

$$A \times A \rightarrow A$$

such that

$$(xy)z = x(yz).$$

A Lie algebra is a vector space $\mathfrak{g}$ with a bilinear bracket

$$[-, -] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$$

satisfying

$$[x, y] = -[y, x],$$

and the Jacobi identity

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

The contrast is important: associative algebras encode composition-like products, while Lie algebras isolate infinitesimal commutation structure.

2.7 Constructed Algebras

Tensor algebra. For a vector space V ,

$$T(V) = \bigoplus_{k=0}^{\infty} V^{\otimes k}.$$

Exterior algebra. The exterior algebra imposes antisymmetry through the relation

$$v \wedge v = 0.$$

Clifford algebra. For a quadratic form g ,

$$uv + vu = 2g(u, v).$$

These constructions are useful because they move from a bare vector space to larger algebras that remember multilinear, antisymmetric, and metric structure.

2.8 Normed and Involutive Algebras

A Banach algebra is an algebra A with a norm $\|\cdot\|$ such that A is complete and

$$\|xy\| \leq \|x\|\|y\|.$$

A C^* -algebra is a Banach algebra equipped with an involution

$$* : A \rightarrow A$$

satisfying

$$(ab)^* = b^*a^*,$$

$$(a^*)^* = a,$$

$$\|a^*a\| = \|a\|^2.$$

3 Analysis

3.1 Functions and Composition

A function is a map

$$f : X \rightarrow Y.$$

Composition is defined by

$$(f \circ g)(x) = f(g(x)).$$

The main point is that analysis starts with maps, but quickly moves to structured maps: differentiable maps, integral operators, variational functionals, and solution operators for differential equations.

3.2 Multi-Index Notation

Let

$$J = (j_1, \dots, j_n) \in \mathbb{N}^n, \quad |J| = \sum_{i=1}^n j_i.$$

For a sufficiently smooth function u ,

$$D_J u = \frac{\partial^{|J|} u}{\partial x_1^{j_1} \cdots \partial x_n^{j_n}}.$$

This notation compresses higher derivatives into a form that scales well when moving from ordinary calculus to PDEs and variational systems.

3.3 Jets and Total Derivatives

Let $u^\alpha(x)$ be unknown fields. The associated jet coordinates are

$$(x^i, u^\alpha, u_J^\alpha).$$

The total derivative in the x^i -direction is

$$D_i = \frac{\partial}{\partial x^i} + \sum_J u_{J,i}^\alpha \frac{\partial}{\partial u_J^\alpha}.$$

This is the right derivative when the variables include not only the fields but also all of their derivatives.

3.4 Integration and Functionals

For a domain $\Omega \subset \mathbb{R}^n$,

$$\int_{\Omega} f(x) \, dx$$

is the integral of a scalar density over Ω .

A variational functional has the form

$$\mathcal{F}[u] = \int_{\Omega} L(x, u, u_J) \, dx,$$

where L is the Lagrangian density. This turns a field u into a scalar quantity and is the basic setup behind the calculus of variations.

3.5 Euler–Lagrange Equation

The stationarity condition for $\mathcal{F}[u]$ is

$$\sum_J (-D)_J \frac{\partial L}{\partial u_J} = 0.$$

This equation is one of the central bridges between analysis and physics: it converts an extremization principle into a differential equation.

3.6 Differential Equations and Boundary Conditions

A general nonlinear differential equation can be written as

$$F(x, u, u_J) = 0.$$

Standard boundary conditions include:

Dirichlet.

$$u = g.$$

Neumann.

$$\frac{\partial u}{\partial n} = g.$$

Robin.

$$au + b \frac{\partial u}{\partial n} = g.$$

3.7 Weak Forms

Instead of solving a differential equation pointwise, one often seeks

$$u \in V$$

such that

$$a(u, v) = \ell(v) \quad \text{for all } v \in V.$$

This formulation is central in finite elements, functional analysis, and modern PDE theory because it weakens differentiability requirements while preserving the core operator structure.

3.8 Banach and Hilbert Spaces

A Banach space is a complete normed vector space with norm

$$\|u\|.$$

A Hilbert space is a complete inner-product space with inner product

$$\langle u, v \rangle.$$

The extra inner-product structure in Hilbert spaces gives orthogonality, projection, and spectral theory in a particularly clean form.

3.9 Operators and Spectral Problems

A linear operator is a map

$$L : V \rightarrow W.$$

The corresponding eigenvalue problem is

$$Lu = \lambda u.$$

This equation organizes much of analysis around invariant modes, frequencies, and stability behavior.

4 Geometry

4.1 Smooth Manifolds

A smooth manifold is a topological space M equipped with an atlas

$$\{(U_i, \phi_i)\},$$

where

$$\phi_i : U_i \rightarrow \mathbb{R}^n$$

are compatible coordinate charts.

The point is that geometry is built by moving between local Euclidean coordinates and globally defined objects that do not depend on a particular chart.

4.2 Tangent, Cotangent, and Tensor Spaces

At each point $p \in M$, the tangent space is

$$T_p M,$$

and the cotangent space is

$$T_p^* M.$$

From these, one builds tensor bundles

$$T_s^r M.$$

These are the natural homes for vector fields, covectors, metrics, curvature, and all other multilinear geometric quantities.

4.3 Differential Forms and Exterior Calculus

A differential k -form is an element

$$\omega \in \Omega^k(M).$$

The exterior derivative is

$$d : \Omega^k(M) \rightarrow \Omega^{k+1}(M), \quad d^2 = 0.$$

The exterior product is

$$\omega \wedge \eta.$$

This package of ideas turns geometry into a calculus of antisymmetric tensors that behaves well under coordinate change.

4.4 Metrics and Connections

A metric is a symmetric $(0, 2)$ -tensor

$$g = g_{ij} \, dx^i \otimes dx^j.$$

A connection differentiates tensor fields. On vector fields, one writes

$$\nabla_X Y,$$

or locally

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k.$$

Connections encode how vectors are transported and compared across nearby points of a manifold.

4.5 Curvature

The curvature operator is

$$R(X, Y)Z.$$

In coordinates, its components are written

$$R^i_{jkl}.$$

The Ricci tensor is obtained by contraction:

$$R_{ij}.$$

The scalar curvature is a second contraction:

$$R = g^{ij} R_{ij}.$$

Together these quantities measure how far the geometry is from being flat.

4.6 Geodesics

A geodesic $\gamma(t)$ satisfies

$$\frac{D\dot{\gamma}}{dt} = 0.$$

This is the intrinsic version of straight-line motion: the acceleration vanishes with respect to the connection, not necessarily in ambient Euclidean coordinates.

4.7 Bundles and Gauge Geometry

A fiber bundle has the form

$$F \hookrightarrow E \rightarrow M.$$

A principal G -bundle is written

$$G \hookrightarrow P \rightarrow M.$$

A connection one-form is denoted by

$$A.$$

Its curvature two-form is

$$F = dA + A \wedge A.$$

The covariant exterior derivative is

$$D = d + A.$$

This notation is the natural language for gauge theory, differential geometry, and many geometric formulations of physics.

4.8 Hodge Theory and Integral Identities

The Hodge star operator is

$$* : \Omega^k(M) \rightarrow \Omega^{n-k}(M).$$

The Laplace–Beltrami operator is

$$\Delta = d\delta + \delta d.$$

Stokes’ theorem closes the circle between local differentiation and global integration:

$$\int_M d\omega = \int_{\partial M} \omega.$$

5 Closing View

One way to read the document is as a progression in which each subject refines the last:

- algebra studies operations and the identities they satisfy,
- analysis studies functions, operators, and variational equations built from those operations,
- geometry studies spaces on which those operators and equations become coordinate-invariant.

The long-term aim is not merely to collect formulas, but to express a style of thinking: start from the primitive objects, choose notation that scales, and let the equations reveal the structure.